

ORF307: Optimization

Precept: #4

Agenda

- Approximate solutions of linear equations with multiple RHS
- Varying the RHS in linearly constrained Least Squares
- Nearest vector with given average
- Multi-objective control (see section 15.2 in Boyd & Vandenberghe book [VMLS])
- Rendezvous (see Python notebook in Companion)

Reminders and Announcements

- Always check Python notebooks in Companion (in directories `lectures/` and `precepts/`)
- Write down the deadline of the current HW:
- Other announcements on blackboard.

Approximate solution of linear equations with multiple RHS's

- Suppose A is invertible
- We seek x such that $Ax \approx b_i$ for $i = 1, 2, \dots, k$

Example:

- Bob proposes a *multi-objective* least squares approach

$$x^{Bob} = \arg \min_x \|Ax - b_1\|_2^2 + \dots + \|Ax - b_k\|_2^2$$

- Alice proposes to first average the b_i 's to get $b = (b_1 + \dots + b_k)/k$ and then choose x by solving the system of linear equations

$$Ax^{Alice} = b$$

- Oscar says that the two approaches sound different, but actually we have:

$$x^{Bob} = x^{Alice}$$

- Is Oscar right?

Solution:

Let's compute Bob's and Alice's solutions, and then try to see if Oscar is right or not.

- Alice's approach

- Bob's approach

- Oscar's claim is

Varying the RHS in *constrained* linear least squares

- Consider the constrained linear least squares problem

$$\begin{aligned}\hat{x} &= \arg \min_x \|Ax - b\|_2^2 \\ &s.t. \ Cx = d\end{aligned}$$

- Assume the KKT matrix is invertible
- Think of $\hat{x}$ as a function of b and d
- Our friend Oscar asserts that $\hat{x}$ is a *linear function* of b and d and hence must have the form

$$\hat{x} = Fb + Gd$$

for some appropriate matrices F and G .

- Is Oscar right? If yes, justify and explain how to find F and G . If not, give a counter-example.

Solution:

Nearest vector with a given average

- Consider the vector $a \in \mathbb{R}^n$ and the scalar $\beta \in \mathbb{R}$
- Find $x \in \mathbb{R}^n$ such that it is closest to vector a among all n -vectors that have average value β .

Solution: